

# Baseline Performance

| Metric     | Value  |
|------------|--------|
| Epochs     | 100    |
| Time       | 2h 10m |
| Final Loss | 0.102  |
| BLEU Score | 0.55   |

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# Tuned Hyperparameters

| Parameter             | Range       |
|-----------------------|-------------|
| Learning Rate         | 1e-5 – 1e-3 |
| Batch Size            | 16, 32, 64  |
| Attention Heads       | 4, 8        |
| Feedforward Dimension | 1024, 2048  |
| Dropout               | 0.1 – 0.4   |

Search Algorithm:

- **OptunaSearch**
- **ASHA Scheduler**

# Best Configuration Found

| Parameter       | Value           |
|-----------------|-----------------|
| Learning Rate   | 0.00072<br>9865 |
| Batch Size      | 32              |
| Attention Heads | 8               |

|                       |         |
|-----------------------|---------|
| Feedforward Dimension | 2048    |
| Dropout               | 0.18409 |
|                       | 3       |

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## Final Model Performance

| Metric     | Value      |
|------------|------------|
| Epochs     | 40         |
| Time       | 20 minutes |
| Final Loss | 0.20       |
| BLEU Score | 0.78       |

### Result:

The tuned model achieved a higher BLEU score while reducing training epochs from 100 → 40.